SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 12

Case Study: Railway Reservation System – Negative Test Scenarios and Test Cases

AIM:

To identify and prepare negative test scenarios and test cases for the Railway Reservation System to verify that the application properly handles invalid inputs, incorrect passenger details, unavailable seats, failed payments, invalid PNR numbers, and unauthorized booking operations while ensuring data security, booking accuracy, and reliable system performance.

1. Problem Description

Negative testing verifies that the system safely handles incomplete searches, invalid dates, unavailable seats, invalid passenger data, failed payments and unauthorized booking access.

2. Functional Requirements

1. Users must search using valid source, destination and journey date.
2. Train schedules and seat availability must be accurate.
3. Passenger details must satisfy required formats and limits.
4. A confirmed booking must have a unique PNR.
5. Ticket confirmation depends on successful payment and booking response.
6. Users may view, retrieve and cancel only authorized bookings.
7. Invalid searches and booking attempts must show clear messages.

3. Negative Test Cases

Negative testing verifies that invalid, incomplete, or unexpected inputs are rejected safely with suitable messages.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Leave source station blank	Search without source	Validation message displayed	Pass
NT02	Select same source and destination	Choose identical stations	Search rejected with suitable message	Pass
NT03	Select a past journey date	Enter a date before today	Past date rejected	Pass
NT04	Book when no seats are available	Select a full train/class	Booking prevented or waitlist option shown	Pass

NT05	Enter invalid passenger age	Enter letters or a negative age	Invalid age rejected	Pass
NT06	Exceed passenger limit	Add more passengers than permitted per booking	Additional passengers rejected	Pass
NT07	Payment failure	Simulate declined payment	Ticket not confirmed; payment status handled safely	Pass
NT08	Enter invalid PNR	Enter a malformed or nonexistent PNR	PNR not found message displayed	Pass
NT09	Cancel another user's ticket	Attempt to access a booking without authorization	Access denied	Pass
NT10	Duplicate booking submission	Submit the same booking request repeatedly	Duplicate booking prevented or clearly handled	Pass

4. Detailed Execution of a Failed Test Case

Example: Select a test case that fails during actual execution. Record the precondition, steps, expected result, and actual result. Do not mark a test as failed unless the defect is observed.

5. Sample Bug Report

Field	**Details**
Bug ID	BR-001
Linked Test Case	Relevant failed negative test case
Severity / Priority	Assign according to impact and urgency
Expected Result	Invalid operation is rejected with a clear message
Actual Result	Application accepts or processes the invalid operation
Suggested Fix	Add server-side validation and repeat checks before final processing
Status	Open – awaiting developer investigation and correction

Conclusion

The negative test cases help verify that the Railway Reservation System meets its requirements and responds appropriately to the tested conditions.